

Surface area and volume change approximation of the lower tongue of the Great Aletsch Glacier using Sentinel-2 and DEM Data

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1. Introduction

Glaciers are important indicators of how climate changes because their surface area and thickness react strongly to changes in temperature and precipitation. Long-term glacier retreat can be observed using satellite imagery, while elevation-change data can be used to estimate how much ice volume has been lost. In this project, the lower tongue of the Great Aletsch Glacier in Switzerland was selected as the study area. This glacier was chosen because its shape is clearly visible in satellite imagery, and the lower tongue can be separated relatively easily from the surrounding mountain terrain.

The main objective of the project was to analyze changes in the surface area of the selected glacier tongue using Sentinel-2 satellite images from three different years: 2015, 2020 and 2025. The analysis was based mainly on multispectral image processing methods, especially spectral indices useful for snow and ice detection. Additionally, a DEM-derived elevation-change raster for the period 2015-2020 was used to approximate the volume change of the glacier tongue.

The combination of Sentinel-2 imagery and DEM-derived elevation change makes it possible to describe the glacier not only by its visible outline, but also by changes in the surface height. This allows the analysis to include both horizontal retreat, expressed as surface area change, and vertical thinning, expressed as approximate volume change. As a result, the project gives a broader view of the changes occurring in the selected part of this glacier.

2. Data

The main data source was Sentinel-2 imagery downloaded from the Copernicus Browser. Images were selected for the same area of interest covering the lower tongue of the Great Aletsch Glacier. Three late-summer dates were used: 2015, 2020 and 2025. Late-summer scenes were chosen because seasonal snow cover is usually lower during this period, which makes the glacier boundary easier to identify.

For each year, raw Sentinel-2 bands were downloaded and used in the analysis. The most important bands were:

- B02 - blue,
- B03 - green,
- B04 - red,
- B08 - near infrared,
- B11 - short-wave infrared,
- B12 - short-wave infrared.

The RGB visualization was created using bands B04, B03 and B02. A false color composition was created using bands B08, B04 and B03. These visualizations helped to visually inspect the glacier, surrounding rock, vegetation, shadows and snow-covered areas.

The DEM-derived part of the project used the file:

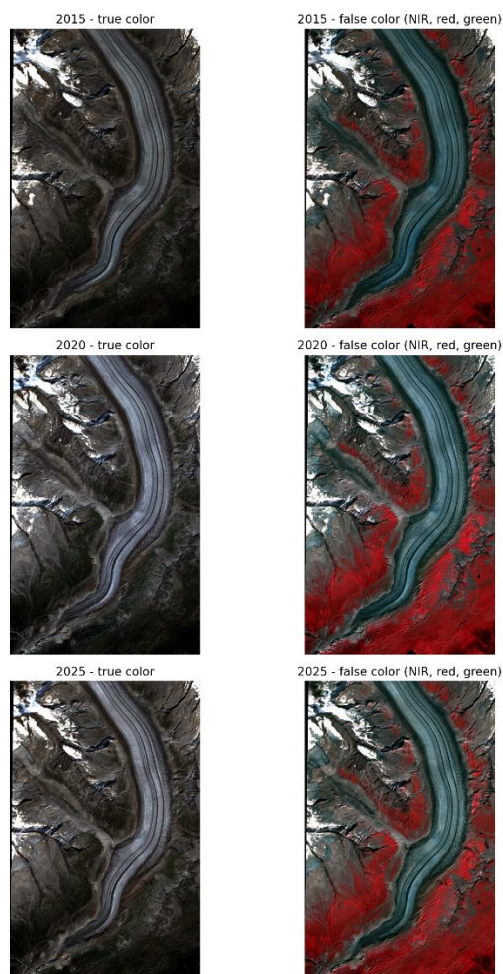
N46E008_2015-01-01_2020-01-01_dhdt.tif

This raster covers the region around 46°N and 8°E, which includes the Great Aletsch Glacier. The file contains *dhdt*, meaning average annual elevation change rate, for the period from 2015-01-01 to 2020-01-01. This dataset was used to estimate the total elevation change and approximate glacier volume change for the selected glacier tongue. The glacier-volume method is based on DEM differencing, where elevation change is calculated from two elevation surfaces and then converted to volume by multiplying by pixel area.

Figure 1. Study area - lower tongue of the Great Aletsch Glacier shown in L1C true color Sentinel-2 composition. Left (29th of August 2015), middle (27th of August 2020), right (26th of August 2025)



Figure 2. True color and false color Sentinel-2 compositions for 2015, 2020 and 2025.



3. Methods

3.1 Spectral indices

The main spectral index used for glacier detection was the Normalized Difference Snow Index:

$$NDSI = (B03 - B11) / (B03 + B11)$$

NDSI is useful because snow and clean glacier ice are usually bright in the green band and darker in the short-wave infrared band. As a result, snow and ice pixels usually have high NDSI values.

To reduce confusion with vegetation, the Normalized Difference Vegetation Index was also calculated as:

$$NDVI = (B08 - B04) / (B08 + B04)$$

Vegetation usually has high NDVI values, while glacier ice and snow should have low or negative NDVI values. Therefore, the final threshold-based glacier mask was created by combining NDSI and NDVI conditions.

3.2 Glacier segmentation

The first segmentation method was a threshold-based method. A pixel was classified as glacier when its NDSI value was high enough and its NDVI value was low enough. This method is simple, interpretable and well suited for detecting bright snow and ice surfaces.

The second method was K-means clustering. This method groups pixels into clusters based on their spectral/index values. It was used as a comparison method to check whether an unsupervised segmentation approach gives similar glacier masks. However, K-means was treated mainly as an additional method, because it can be less stable between dates when illumination, seasonal snow and shadows change.

After segmentation, small isolated objects and small holes were removed using morphological operations. This improved the visual quality of the masks and reduced noise in the calculated glacier area.

Figure 3. NDSI and NDVI maps for 2015, 2020 and 2025.

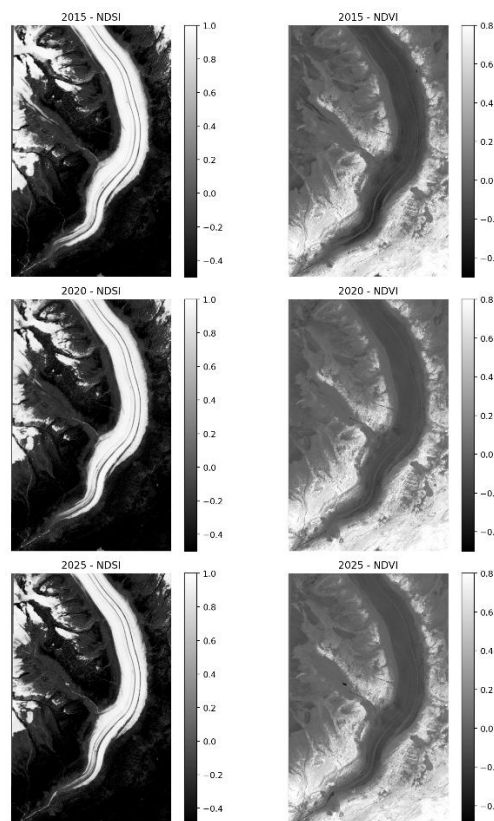
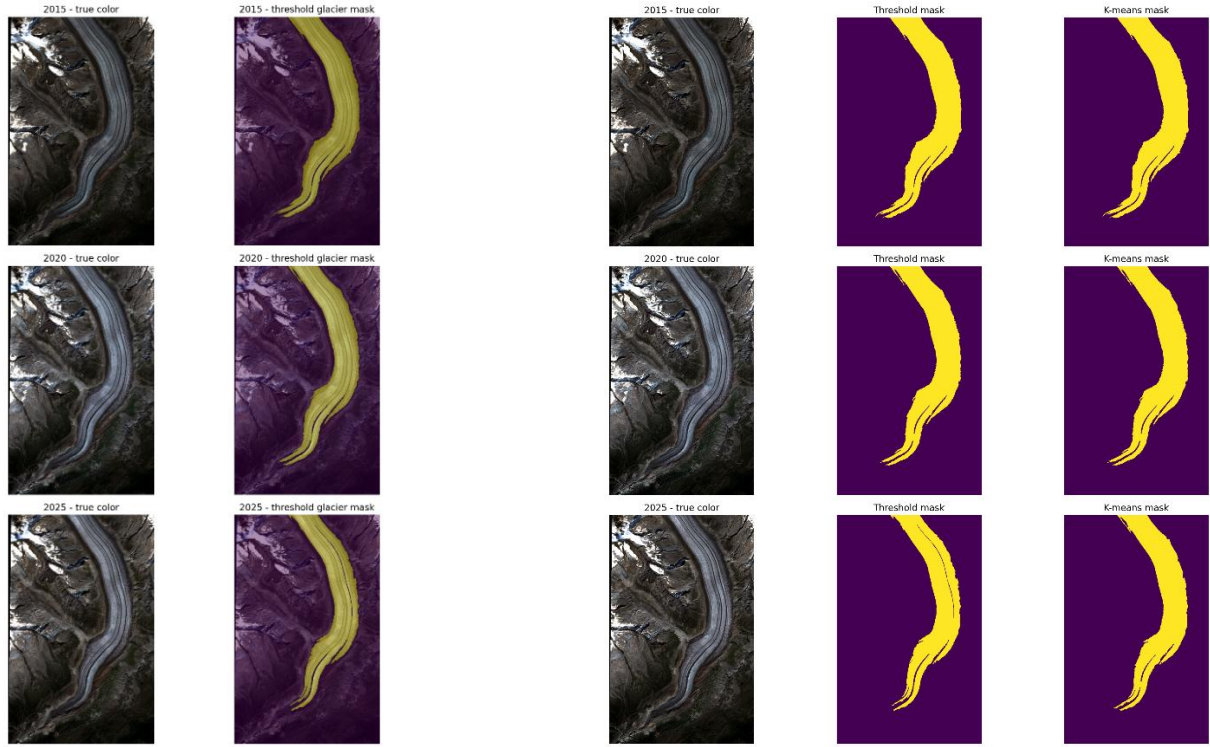


Figure 4. Threshold-based glacier masks for 2015, 2020 and 2025. (left)

Figure 5. Comparison of threshold-based and K-means glacier segmentation. (right)



3.3 Surface area calculation

The surface area of the glacier tongue was calculated from the number of glacier pixels in the final mask. Because the Sentinel-2 bands were processed at 10 m resolution, one pixel corresponds to:

$$10\text{ m} \times 10\text{ m} = 100\text{ m}^2$$

The glacier surface area was calculated as:

$$\text{area} = \text{number of glacier pixels} \times 100\text{ m}^2$$

The result was then converted to km^2 . This calculation was repeated for 2015, 2020 and 2025, allowing comparison of the glacier surface area between dates.

3.4 DEM-derived elevation and volume change

The DEM-derived file contained average annual elevation change rate, $dhdt$, for 2015-2020. To obtain total elevation change for this five-year period, the raster values were multiplied by 5:

$$dH = dhdt \cdot 5$$

The volume change was then approximated by multiplying the total elevation change by the DEM pixel area and summing over the selected glacier mask:

$$volume\ change = \sum dH \times pixel\ area$$

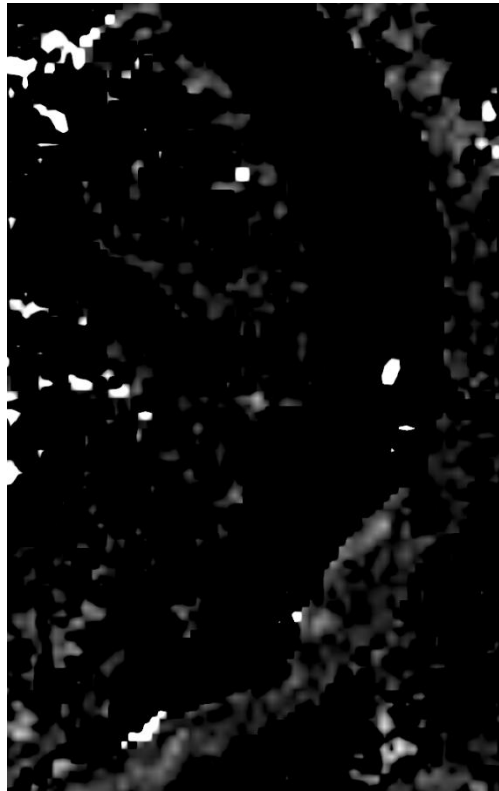
For a 100 m DEM raster, the pixel area is:

$$100\ m \times 100\ m = 10000\ m^2$$

This volume estimate is approximate because it depends on the quality of the glacier mask, the resolution of the DEM-derived raster and the selected glacier area.

However, it follows the standard principle of converting elevation change into volume change.

Figure 6. DEM-derived elevation change map for 2015-2020.



4. Results

The true color and false color visualizations showed that the lower tongue of the Great Aletsch Glacier is clearly visible in all three analyzed years. The glacier has a long curved shape and is surrounded mostly by rock, mountain slopes and vegetation in the lower part. This made the selected area suitable for image segmentation.

The NDSI maps highlighted the main snow and ice-covered areas. High NDSI values corresponded mainly to glacier ice and snow patches, while lower values appeared over rock, vegetation and terrain with shadows. The threshold-based method produced relatively stable masks for all three dates. The glacier tongue was clearly detected, although some difficulties appeared near debris-covered or shadowed parts of the glacier.

The calculated glacier surface areas showed a decreasing trend for the selected lower glacier tongue. The threshold-based result was treated as the main result because it produced a more stable trend between the dates. K-means segmentation was useful for comparison, but its result for 2020 was less consistent, probably because the clusters were affected by differences in brightness, snow patches or some shadows.

Figure 7. Estimated glacier surface area in 2015, 2020 and 2025.

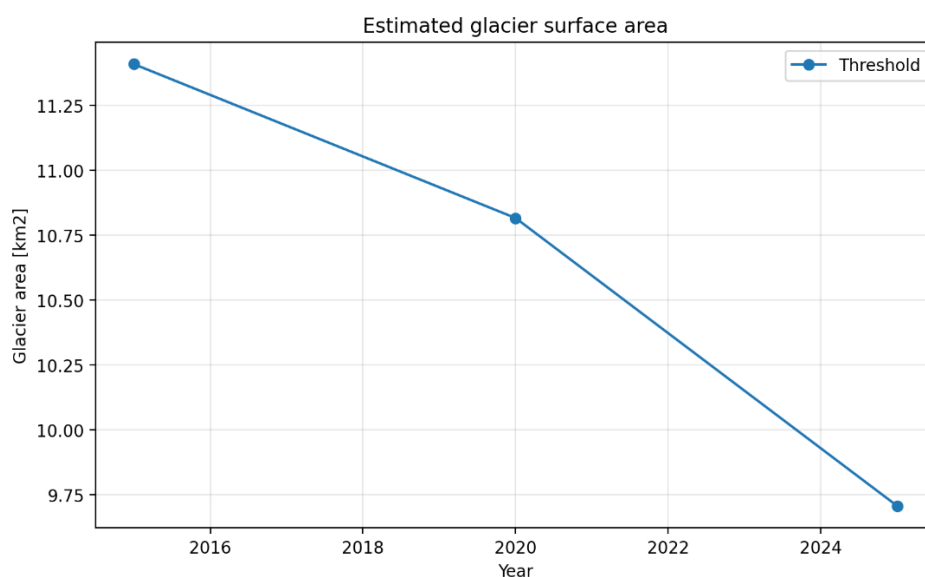
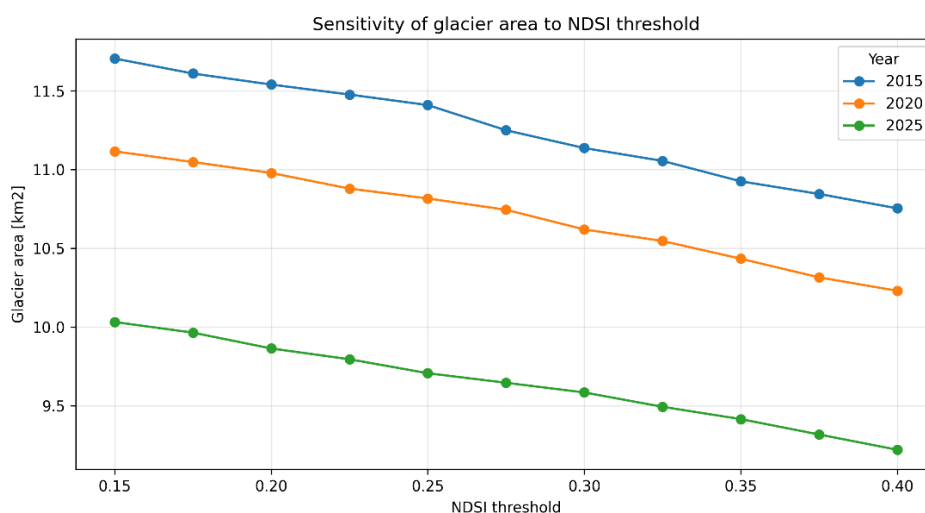


Figure 8. Sensitivity of glacier area estimation to different NDSI threshold values.



The main threshold-based result was calculated using $\text{NDSI} > 0.25$ together with the NDVI condition. The sensitivity test was added to check how much the calculated area changes when the NDSI threshold is modified.

The threshold sensitivity plot shows that the estimated glacier area decreases when the NDSI threshold is increased. This is expected, because stricter threshold values classify fewer pixels as glacier. However, the order of the years stays the same for all tested thresholds: 2015 has the largest glacier area, 2020 is smaller and 2025 is the smallest. This means that the decreasing trend is stable and is not only caused by one selected threshold value.

Table 1. Glacier surface area calculated from the threshold-based mask.

Year	Glacier area [km^2]
2015	11.40
2020	10.81
2025	9.70

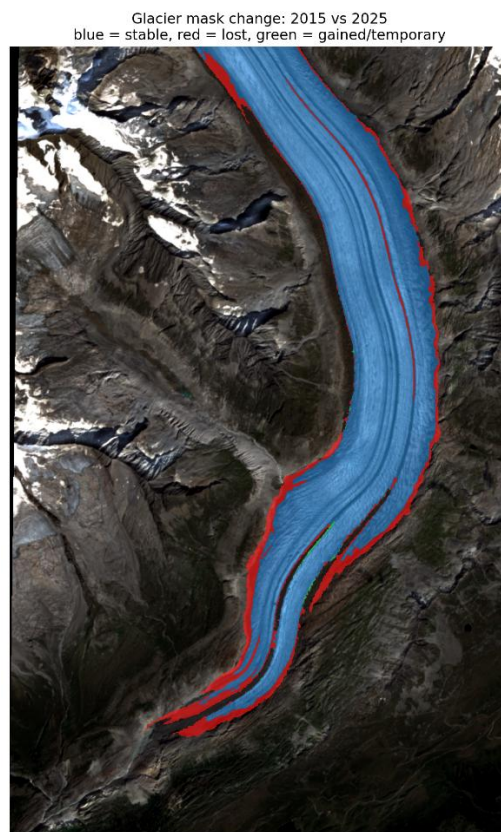
Table 2. Glacier surface area change between dates.

Period	Area change [km^2]	Relative change [%]
2015-2020	-0.59	-5.17
2015-2025	-1.70	-14.91

Between 2015 and 2025, the selected glacier tongue area decreased by 1.70 km^2 , which corresponds to approximately 14.91% of the 2015 area.

The change map between 2015 and 2025 showed the spatial distribution of glacier retreat. The most important changes were expected near the lower tongue and glacier margins, where the glacier boundary is most sensitive to melting and retreat.

Figure 9. Glacier change map between 2015 and 2025 showing stable glacier area and lost glacier area.

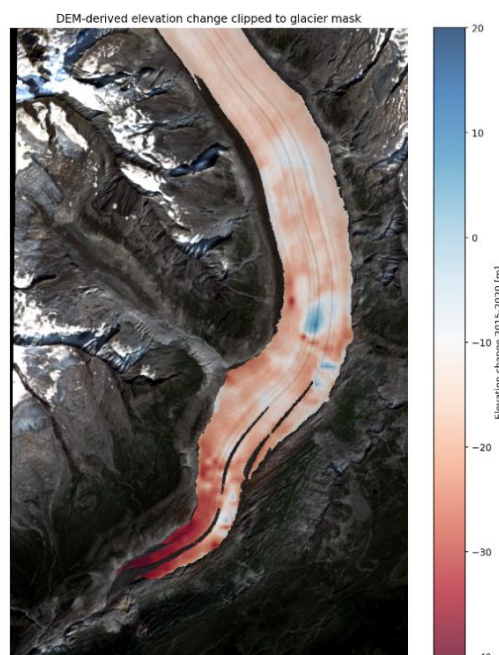


The DEM-derived elevation-change raster showed surface elevation changes for the 2015-2020 period. Negative values indicate thinning, while positive values indicate local thickening or possible uncertainty/noise. After clipping the raster to the glacier mask, the total elevation change was converted to an approximate volume change. This provided an additional quantitative estimate of glacier change, complementary to the surface area analysis.

Table 3. DEM-derived elevation and volume change for 2015-2020.

Quantity	Value
Mean elevation change [m]	-17.76
Volume change [m^3]	-202633792.0
Volume change [km^3]	-0.202634

Figure 10. DEM-derived elevation change map clipped to the glacier mask 2015-2020



5. Conclusions

The lower tongue of the Great Aletsch Glacier was successfully analyzed using Sentinel-2 imagery from 2015, 2020 and 2025. The glacier surface was segmented using spectral indices, mainly NDSI, with NDVI used to reject vegetation. The calculated glacier masks allowed to estimate and compare glacier surface area between the selected dates.

The results showed a general decrease in the glacier surface area of the selected tongue. The threshold-based segmentation method gave the clearest and most reliable results. K-means was useful as a comparison method, but it was less reliable for this dataset because the clustering result varied between dates.

The DEM-derived *dhd* raster for 2015-2020 was used to estimate elevation and volume change. This part of the analysis showed how two-dimensional glacier masks can be combined with elevation-change data to obtain an approximate three-dimensional estimate of glacier loss.

Overall, the project showed that the Sentinel-2 images are useful for monitoring glacier surface area change, while DEM-derived elevation change data can extend the analysis towards approximate volume change. The most important limitations were seasonal snow, glacier ice covered in debris, shadows and the lower resolution of the DEM-derived raster.